

Telecom Customer Churn: Key Insights

Exploratory analysis of 504,000 customers | Python, pandas, Seaborn | GinishTech

504,000
customers analysed

34.3%
overall churn rate

49.1%
month-to-month churn

62.8%
month-to-month + Fiber churn

1. Contract type is the strongest driver

Customers on month-to-month plans churn at about 49%, versus 19.6% on one-year and 11.8% on two-year contracts. Locking customers into longer terms is the biggest retention lever in this data.

Contract	Churn %
Month-to-month	49.1
One year	19.6
Two year	11.8

2. Fiber customers are the highest risk

Fiber churns at 45.2% (DSL 27.8%, no internet 19.0%). Combined with contract type, month-to-month Fiber customers churn at **62.8%**, the riskiest segment, while two-year contract customers with no internet churn at only 4.4%.

Contract	DSL	Fiber	No internet
Month-to-month	41.3	62.8	29.3
One year	13.8	28.6	8.2
Two year	7.8	17.7	4.4

3. Support calls and tenure are early-warning signals

Churn climbs steeply with support calls: 19.7% for customers with none, 49.9% for 3-4 calls and **64.4% for 5+ calls**. New customers are also fragile: 50.1% of those in their first 12 months churn, falling to 22.3% after four years. Churned customers also pay more (avg Rs. 888 vs Rs. 773 per month).

Support calls	Churn %	Tenure (months)	Churn %
0	19.7	0-12	50.1
1-2	32.2	13-24	43.3
3-4	49.9	25-48	34.0
5+	64.4	49+	22.3

4. What does NOT matter

Age group (all about 34%), gender (34.3% vs 34.2%), payment method (33.7% to 38.8%) and paperless billing (34.2% vs 34.3%) show little to no effect on churn. City differences are small (roughly 32% to 37%), and part of that gap is a data issue: city names appear with inconsistent capitalisation (for example "Pune" and "pune") and should be standardised.

Recommendations

- Offer discounts or perks to move month-to-month customers onto 12-month plans, starting with Fiber users.
- Flag customers with 3+ support calls for proactive outreach and faster issue resolution.
- Build an onboarding and check-in programme for the first 12 months, when churn is highest.
- Review Fiber service quality and pricing, since it combines the highest churn with the highest charges.

Data notes

Figures are computed on the raw dataset of 504,000 rows. About 3% of rows have missing values in age, city, monthly or total charges, and roughly 4,000 customer IDs are duplicated. These findings show associations, not proven causes.

